

A disproof of Conjecture 00000007993

The singular set of the obstacle problem can have dimension $n - 1$

1 The conjecture

Conjecture 00000007993 (English text) reads: “Definition: The stratification of free boundary regularity for the obstacle problem: the proportions of regular and singular parts of the free boundary $\text{FB}(u)$ of solutions of obstacle problems of $u = \max(w, 0)$ type. Conjecture: The regular part of FB is an $(n-1)$ -dimensional C^∞ surface, relatively open and dense in FB ; the Hausdorff dimension of the singular part is at most $n-3$ for n at least 3, and at most 1 for $n = 2$ (degenerating to points); and the codimension threshold of the regular-singular decomposition is given by the tangent cone classification: [...]”. The Chinese text says the same (the regular part is relatively open and dense in FB ; the singular part has Hausdorff dimension at most $n - 3$ for $n \geq 3$).

The conjecture is a conjunction. We refute two of its clauses:

(D) for $n \geq 3$, every solution has $\dim_{\mathcal{H}} \text{Sing}(u) \leq n - 3$;

(R) the regular part is dense in the free boundary, i.e. $\text{FB}(u) \subseteq \overline{\text{Reg}(u)}$.

Refuting either clause refutes the conjecture. We do not address the $n = 2$ clause (our example gives dimension 1 there, which is allowed) or the tangent-cone clause.

2 Reading and definitions

The equation. “Obstacle problem” is read as the classical (normalized) obstacle problem

$$\Delta u = \chi_{\{u > 0\}}, \quad u \geq 0 \quad \text{in } \Omega,$$

which is how Figalli, Ros-Oton and Serra write it (arXiv:1912.00714, Section 3, equation (3.1)). Its solutions satisfy $u = \max(u, 0)$; this is our reading of the garbled phrase “ $u = \max(w, 0)$ type”. Our witness is also $\max(w, 0)$ for $w = x_1^2/2$, which solves $\Delta w = 1$.

Solution (Lean: `IsObstacleSolution`). Here $\Omega \subseteq \mathbb{R}^n$ is open and $u : \mathbb{R}^n \rightarrow \mathbb{R}$. We require that u is C^∞ , that $u \geq 0$ on Ω , that $\Delta u = 1$ at every point of $\{u > 0\} \cap \Omega$, and that $\Delta u = \chi_{\{u > 0\}}$ holds Lebesgue-almost everywhere in Ω . Solutions are usually only $C^{1,1}$, and for them the equation holds a.e.; for smooth u the a.e. equation is the same as the distributional one. This definition asks for more than the standard one. A stronger definition can only shrink the set of solutions, which weakens the universal claims (D) and (R). Refuting them is therefore at least as strong as refuting the standard versions. Δ is Mathlib’s Laplacian $\Delta u(x) = \sum_i D^2 u(x)[e_i, e_i]$ over an orthonormal basis.

Free boundary. $\text{FB}(u) = \Omega \cap \partial\{u > 0\}$ (Lean: `freeBoundary`).

Regular and singular points. We use the definitions of Figalli, Ros-Oton and Serra (arXiv:1912.00714, Section 3), quoted verbatim: “Regular points: $x_o \in \partial\{u > 0\}$ is a regular point if $\lim_{r \rightarrow 0} r^{-2}u(x_o + rx) = \frac{1}{2}(\max\{0, e \cdot x\})^2$ for some $e \in \mathbb{S}^{n-1}$. Singular points: $x_o \in \partial\{u > 0\}$ is a singular point if $p_{2,x_o}(x) := \lim_{r \rightarrow 0} r^{-2}u(x_o + rx)$ exists and p_{2,x_o} is a quadratic polynomial belonging to the set $\mathcal{P} := \{\text{convex 2-homogeneous polynomials } p \text{ with } \Delta p \equiv 1\}$.” In Lean the rescaling is $u_{x_o,r}(x) = u(x_o + rx)/r^2$ (**rescale**) and $r \rightarrow 0^+$.

- **IsRegularPt:** $x_o \in \text{FB}$ and $u_{x_o,r}(x) \rightarrow \frac{1}{2}(\max\{0, \langle e, x \rangle\})^2$ *pointwise* for some unit e . Pointwise convergence is the weakest mode, so “not regular” is the strongest form of the negation.
- **IsSingularPt:** $x_o \in \text{FB}$ and $u_{x_o,r} \rightarrow p$ *locally uniformly* for some $p \in \mathcal{P}$ (**polyClass**: $p(x) = Q(x, x)$ for a continuous bilinear Q , p convex on $\mathbb{R}^n$, $\Delta p \equiv 1$).

$\text{Reg}(u)$ and $\text{Sing}(u)$ are the sets of such points. Hausdorff dimension is Mathlib’s **dimH** on $\mathbb{R}^n$ with the Euclidean metric (**EuclideanSpace** $\mathbb{R}$ (**Fin** n)). Coordinates are indexed from 0 in Lean, so x_1 is **x 0**.

3 The counterexample

Let $n \geq 3$, $\Omega = B_1$ the open unit ball, and $u_0(x) = x_1^2/2$.

Lemma 1 (**laplacian_u0**, **isObstacleSolution_u0**). u_0 is C^∞ and nonnegative, and $\Delta u_0 \equiv 1$. Hence u_0 is a solution in every open set Ω .

Proof. $Du_0(x) = x_1 dx_1$ and $D^2u_0(x)[v, w] = v_1 w_1$, so $\Delta u_0 = \sum_i (e_i)_1^2 = 1$. The set $\{u_0 = 0\} = \{x_1 = 0\}$ is a proper linear subspace, so it has Lebesgue measure 0 (Mathlib **Measure.addHaar_submodule**). Hence $\chi_{\{u_0 > 0\}} = 1 = \Delta u_0$ a.e. $\square$

Lemma 2 (**freeBoundary_u0**, **contact_density_zero**). $\text{FB}(u_0) = \Omega \cap \{x_1 = 0\}$. The contact set $\{u_0 = 0\}$ has measure zero, so its density $|\{u_0 = 0\} \cap B_r(x_o)|/|B_r(x_o)|$ is 0 for every x_o and every $r > 0$ (for $r \leq 0$ the ball is empty; Lean’s division in $[0, \infty]$ then also gives 0, but only $r > 0$ matters for the limit $r \rightarrow 0^+$).

Proof. $\{u_0 > 0\}$ is the complement of the hyperplane $H = \{x_1 = 0\}$. H is closed with empty interior, since a proper subspace with nonempty interior would be everything. So $\partial\{u_0 > 0\} = \partial H = H$. $\square$

Lemma 3 (**rescale_u0**, **singular_u0**, **not_regular_u0**). Every point of $\text{FB}(u_0)$ is singular and none is regular.

Proof. If $(x_o)_1 = 0$ and $r \neq 0$, then $u_0(x_o + rx)/r^2 = (rx_1)^2/(2r^2) = u_0(x)$. So every rescaling equals u_0 , and the limit exists in every sense and is u_0 . Moreover $u_0 \in \mathcal{P}$: take $Q(x, y) = \frac{1}{2}x_1 y_1$. Then u_0 is convex because $t \mapsto t^2$ is, and $\Delta u_0 = 1$. Hence every free-boundary point is singular. Now suppose some point were regular with unit vector e . By uniqueness of limits, $u_0(x) = \frac{1}{2}(\max\{0, \langle e, x \rangle\})^2$ for all x . At $x = -e$ this gives $e_1^2/2 = 0$, so $e_1 = 0$. At $x = e_1^{\text{basis}} = (1, 0, \dots, 0)$ it then gives $\frac{1}{2} = \frac{1}{2}(\max\{0, e_1\})^2 = 0$, a contradiction. $\square$

Lemma 4 (**finrank_plane**, **dimH_inter_plane**). If Ω is open and meets H , then $\dim_{\mathcal{H}}(\Omega \cap H) = n - 1$.

Proof. $H = \ker(x \mapsto x_1)$ has dimension $n - 1$ by rank-nullity. The inclusion $H \rightarrow \mathbb{R}^n$ is an isometry, which preserves $\dim_{\mathcal{H}}$ (**Isometry.dimH_image**). The preimage of Ω in H is a nonempty open set, and a set with nonempty interior in a finite-dimensional real normed space has Hausdorff dimension equal to the vector-space dimension (**Real.dimH_of_nonempty_interior**). $\square$

Theorem 5 (witness, family, not_singDimBound, not_regDenseInFB, not_conjecture). *For every $n \geq 3$, $u_0 = x_1^2/2$ solves the obstacle problem in $B_1 \subseteq \mathbb{R}^n$. It has $\text{Reg}(u_0) = \emptyset$ and $\text{Sing}(u_0) = \text{FB}(u_0) = \text{FB}(u_0) \setminus \text{Reg}(u_0) = B_1 \cap \{x_1 = 0\}$, and $\dim_{\mathcal{H}} \text{Sing}(u_0) = n - 1 > n - 3$. Hence clause (D) is false. Since $0 \in \text{FB}(u_0)$ and $\text{Reg}(u_0) = \emptyset$, clause (R) is false as well.*

Remark 6. The example is degenerate: the contact set $\{u_0 = 0\}$ is a hyperplane with empty interior. It is nevertheless a solution under the cited definition. On B_1 it is also the minimiser of $\int(\frac{1}{2}|\nabla v|^2 + v)$ over $v \geq 0$ with the same boundary values, since the unconstrained minimiser (with $\Delta v = 1$) is already nonnegative (this remark is not formalized). The literature records that the singular set can be this large. Figalli, Ros-Oton and Serra (arXiv:1912.00714, abstract) write: “Explicit examples show that the singular set could be in general $(n - 1)$ -dimensional—that is, as large as the regular set. Our main result establishes that, generically, the singular set has zero $\mathcal{H}^{n-4}$ measure [...]” So a codimension-type bound on Sing holds only generically, and the conjecture states it for all solutions.

4 Lean formalization

The file `lean/Conjecture7993/Basic.lean` (311 lines, only `import Mathlib`, Lean v4.33.1 with Mathlib v4.33.1) proves:

```
def SingDimBound : Prop :=
  forall n : Nat, 3 <= n -> forall Omega : Set (E n), IsOpen Omega ->
    forall u : E n -> Real, IsObstacleSolution Omega u ->
      dimH (singularSet Omega u) <= ((n - 3 : Nat) : ENNReal)
def RegDenseInFB : Prop :=
  forall n : Nat, 2 <= n -> forall Omega : Set (E n), IsOpen Omega ->
    forall u : E n -> Real, IsObstacleSolution Omega u ->
      freeBoundary Omega u <= closure (regularSet Omega u)
theorem family (n : Nat) (hn : 3 <= n) :
  exists u : E n -> Real, (forall x, u x = (x < 0, _>) ^ 2 / 2) /\
    IsObstacleSolution (ball 0 1) u /\ regularSet (ball 0 1) u = {} /\
    dimH (singularSet (ball 0 1) u) = ((n - 1 : Nat) : ENNReal) /\
    ((n - 3 : Nat) : ENNReal) < dimH (singularSet (ball 0 1) u)
theorem not_singDimBound : Not SingDimBound
theorem not_regDenseInFB : Not RegDenseInFB
theorem not_conjecture : Not (SingDimBound /\ RegDenseInFB)
```

(The Lean source uses the Unicode symbols; `E n` is `EuclideanSpace R (Fin n)`.) `witness` additionally records the free-boundary description, `Sing = FB \ Reg`, and the zero contact-set density.

Axiom audit. `lake build` succeeds. `#print axioms` for `not_conjecture`, `not_singDimBound`, `not_regDenseInFB`, `family` and `witness` reports exactly `[propxt, Classical.choice, Quot.sound]`. There is no `sorry`, no `native_decide` and no added axiom.

Sources

- A. Figalli, X. Ros-Oton, J. Serra, *Generic regularity of free boundaries for the obstacle problem*, arXiv:1912.00714 (equation (3.1), the definitions of regular and singular points in Section 3, and the abstract).